

MAINS MASTER PROGRAM (MMP) 2024

ORIENTATION AND DEMO LECTURE

SCIENCE AND TECHNOLOGY-1

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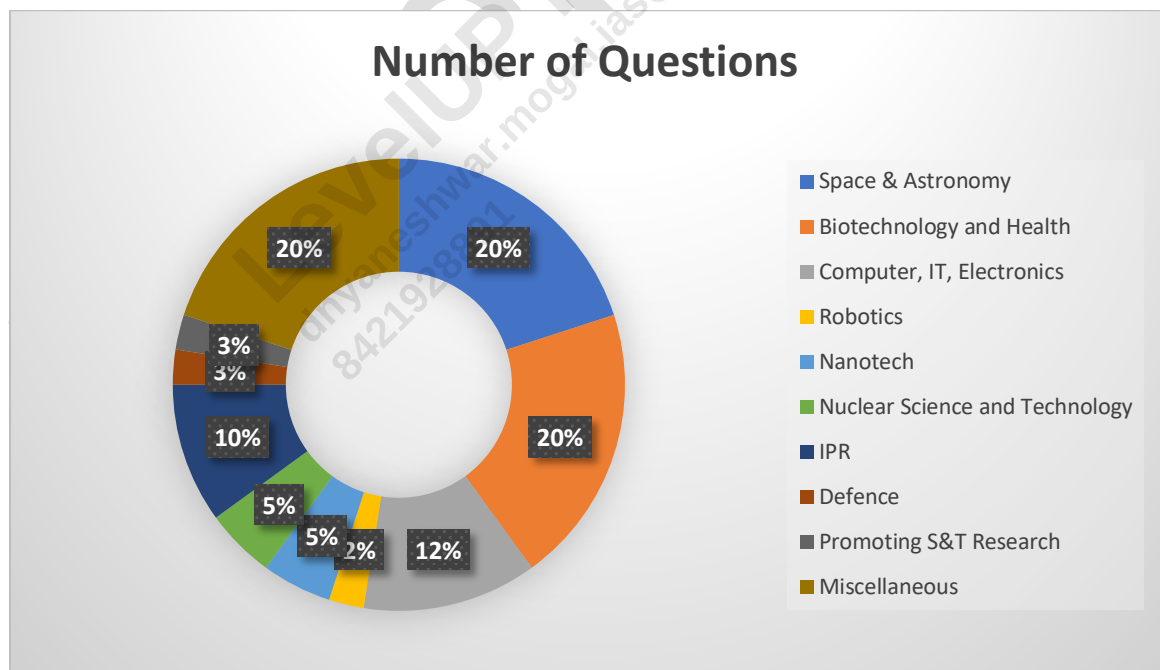
2. SYLLABUS

1. Science and Technology - developments and their applications and effects in everyday life Achievements of Indians in science & technology; indigenization of technology and developing new technology.
2. Awareness in the fields of IT, Space, Computers, robotics, Nanotechnology, biotechnology and issues relating to intellectual property rights.

3. PYQ ANALYSIS

1) DISTRIBUTION OF QUESTIONS FROM VARIOUS AREAS IN LAST 11 YEARS (2013-2023)

Topic	Number of Questions
Space & Astronomy	8
Biotechnology and Health	8
Computer, IT, Electronics	5
Robotics	1
Nanotech	2
Nuclear Science and Technology	2
IPR	4
Defence	1
Promoting S&T Research	1
Miscellaneous	8
Total	40



2) SPACE

1. What is the **main task of India's third moon mission** which could not be achieved in its earlier mission? List the countries that have achieved this task. Introduce the subsystems in the spacecraft launched and explain the role of the 'Virtual Launch Control Center' at the Vikram Sarabhai Space Center which contributed to the successful launch from Sriharikota. [250 words, 15 marks, Mains 2023]
2. Launched on 25th Dec 2021, **James Web Space Telescope** has been much in the news since then. What are its unique features which make it superior to its predecessor space telescopes. What are the key goals of this mission? What potential benefits does it hold for the human race? [250 words, 15 marks, Mains 2022]
3. What is India's plan to **have its own space station** and how will it benefit our space programme? [10 marks, 150 words, Mains 2019]
4. Why is **Indian Regional Navigational Satellite System (IRNSS)** needed? How does it help in navigation? (150 words, 10 marks, Mains 2018 [GS1])
5. India has achieved remarkable successes in unmanned space missions including the Chandrayaan and Mars Orbiter Mission, but has not ventured into **manned space mission**, both in terms of technology and logistics? Explain critically. [150 words, 10 marks, Mains 2017]
6. How does the **Juno Mission of NASA** help to understand the origin and evolution of the Earth? [150 words, 10 marks, Mains 2017 [GS1]]
7. Discuss India's **achievements in the field of Space Science and Technology**. How the application of this technology has helped India in its **socio-economic development**? [Mains 2016]
8. What do you understand by **'Standard Positioning Systems'** and 'Protection Positioning Systems' in the GPS era? Discuss the advantages India perceives from its ambitious IRNSS programme employing just seven satellites. [Mains 2015]

3) BIOTECHNOLOGY AND HEALTH

1. The increase in life expectancy in the country has led to newer health challenges in the community. What are those challenges and what steps need to be taken to meet them? [150 words, 10 marks, Mains 2022]
2. What is the basic principle behind vaccine development? How do vaccines work? What approaches were adopted by the Indian vaccine manufacturers to produce COVID-19 vaccines? [15 marks, 150 words, Mains 2022]
3. What are the research and developmental achievements in applied biotechnology? How will these achievements help to uplift the poorer sections of society? [250 words, 15 marks, Mains 2021]
4. How can biotechnology improve the living standards of farmers [15 marks, 250 words, Mains 2019]
5. Why is there so much activity in the field of biotechnology in our country? How has this activity benefitted the field of Biopharma? [15 marks, 250 words][Mains 2018]

6. Stem cell therapy is gaining popularity in India to treat a wide variety of medical conditions including leukemia, Thalassemia, damaged cornea and several burns. Describe briefly what stem cell therapy is and what advantages it has over other treatments? (150 words, 10 marks)[Mains 2017]
7. Can overuse and free availability of antibiotics without doctor's prescription, be contributors to the emergence of drug-resistant disease in India? What is the available mechanism for monitoring and control? Critically discuss the various issues involved. [Mains 2014]
8. What do you understand by Fixed-Dose drug Combinations (FDCs)? Discuss their merits and demerits. [Mains 2013]

4) COMPUTER, IT & ELECTRONICS

1. Introduce the Concept of Artificial Intelligence (AI). How does AI help clinical diagnosis? Do you perceive any threat to privacy of the individual in the use of Artificial Intelligence in healthcare [Answer in 150 words, 10 marks, Mains 2023]
2. The Nobel Prize in Physics in 2014 was jointly awarded to Akasaki, Amano, and Nakamura for the invention of Blue LEDs in 1990s. How has this invention impacted the everyday life of human beings? (Answer in 250 words) (Mains 2021)
3. Discuss the advantage and security implications of cloud hosting of servers vis-a-vis in-house machine-based hosting for government businesses. [Mains 2015]
4. What is a digital signature? What does its authentication mean? Give various salient built-in features of a digital signature. [Mains 2013]
5. How does 3D printing technology work? List out the advantages and disadvantages of the technology. [Mains 2013]

5) ROBOTICS

1. What are the areas of prohibitive labour that can be sustainably managed by robots? Discuss the initiatives that can propel research in premier research institutes for substantive and gainful innovation.

6) NANOTECH

1. What do you understand by nanotechnology and how is it helping in health sector? [10 marks, 150 words, Mains 2020]
2. Why is nanotechnology one of the key technologies of the 21st century? Describe the salient features of Indian Government's Mission on Nanoscience and Technology and the scope of its application in the development process of the country (2016)

7) NUCLEAR SCIENCE AND TECHNOLOGY

1. With growing energy needs should India keep on extending its nuclear energy program? Discuss the facts and fears associated with nuclear energy. [Mains 2018]
2. Give an account of the growth and development of nuclear science and technology in India. What is the advantage of fast breeder reactor programme in India? (250 words, 15 marks, Mains 2017)

8) INTELLECTUAL PROPERTY RIGHTS

- i) How is Government of India protecting traditional knowledge of medicine from patenting by pharmaceutical companies? [Mains 2019]
- ii) India's Traditional Knowledge Digital Library (TKDL) which has a database containing formatted information on more than 2 million medicinal formulations is proving a powerful weapon in the country's fight against erroneous patents. Discuss the pros and cons of making this database publicly available under open-source licensing [Mains 2015]
- iii) In a globalized world, **intellectual property rights** assume significance and are a source of litigation. Broadly distinguish between the terms – **copyrights, patents and trade secret** [Main 2014]
- iv) Bring out the circumstances in 2005 which forced amendment to section 3(d) in the Indian Patent Law, 1970. Discuss how it was utilized by Supreme Court in its judgment rejecting Novartis patent applications for "Glivec". Discuss briefly the pros and cons of the decision? [12.5 marks, 200 words] [Mains 2013]

9) DEFENCE

1. How is the S-400 air defence system technically superior to any other system presently available in the world? (Answer in 150 words)[Mains 2021]

10) PROMOTING S&T RESEARCH IN INDIA

1. Scientific research in Indian universities is declining, because a career in science is not as attractive as our business operations, engineering or administration, and the universities are becoming consumer-oriented. Critically comment. [Mains 2014]

11) MISCELLANEOUS

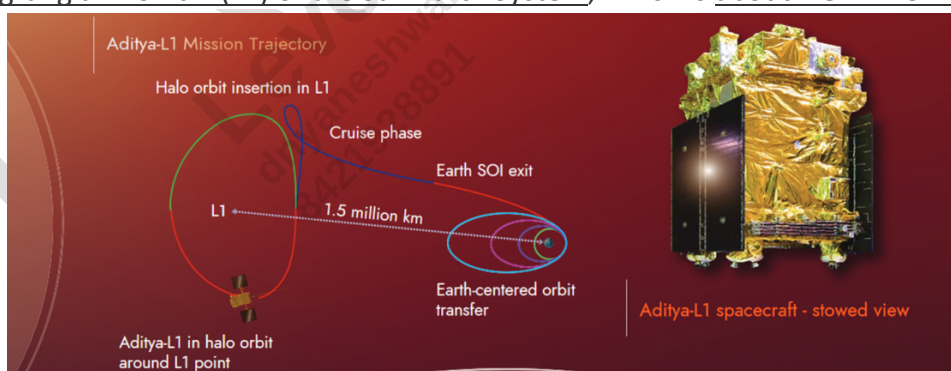
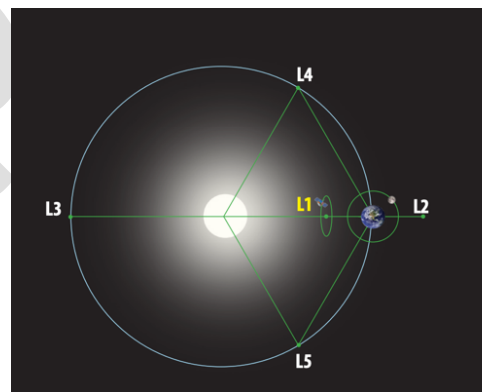
1. Discuss several ways in which micro-organisms can help in meeting the current fuel shortage [10 marks, 150 words, Mains 2023]

2. Each year a large amount of plant material, cellulose, is deposited on the surface of Planet Earth. What are the natural processes this cellulose undergoes before yielding carbondioxide, water and other end products? [10 marks, 150 words, Mains 2022]
3. How is science interwoven deeply in our lives? What are the striking changes in agriculture triggered off by science-based technologies [Mains 2020]
4. Covid-19 pandemic has caused unprecedented devastation worldwide. However, technological advancements are being availed readily to win over the crisis. Give an account of how technology was sought to aid the management of pandemic [Mains 2020]
5. How was India benefitted from the contributions of Sir M. Visvesvaraya and Dr. M. S. Swaminathan in the fields of water engineering and agricultural science respectively? [10 marks, 150 words, Mains 0219]
6. Discuss the work of 'Bose-Einstein Statistics' done by Prof. Satyendra Nath Bose and show how it revolutionized the field of Physics. [Mains 2018]
7. What is allelopathy? Discuss its role in major cropping systems of irrigated agriculture [Main 2016]
8. What is an FRP composite material? How are they manufactured? Discuss their application in the aviation and automobile industries. [Mains 2013]

4. SOME DEMO TOPICS

1) ADITYA-L1

- **Practice Question:** What are the main science objectives of India's Aditya-L1 mission. Introduce the subsystems in the spacecraft launched. [10 marks, 150 words]
- **Introduction:** ADITYA L1 is India's first observatory class space based solar mission. It was launched onboard PSLV-C57 in Sep 2023 and reached L1 point 127 days later in Jan 2024.
- **The major science objectives** of the mission in its five years of mission life are to **understand**:
 - The coronal heating and solar wind acceleration;
 - The initiation of Coronal mass ejection (CME), flares, and near earth space weather;
 - The coupling and dynamics of the solar atmosphere;
 - The solar wind distribution and temperature anisotropy
- **What is the need of studying sun from space and specifically from Lagrangian point 1 (L1):**
 - **Significance of studying Sun from Space:** Radiations which don't reach earth due to earth's magnetic field and atmosphere can be studied.
 - **Significance of studying sun from L1:** A Satellite placed in the halo orbit around the Lagrangian point 1 (L1) of the Sun-Earth system has the major advantage of continuously viewing the Sun without any occultation/ eclipses.
- **What Trajectory ADITYA-L1 followed to reach Lagrangian Point-1:** Through various orbit raising manoeuvres and cruise phase, it was placed in a halo orbit around the Lagrangian Point-1 (L1) of the Sun Earth System, which is about 1.5 million km from the Earth.



The path Aditya-L1 will take to get to L1. | Photo Credit: ISRO

- **Aditya-L1 went with 7 Payloads** (4 remote sensing payloads and 3 in-situ payloads)
 - » **Remote Sensing Payload:**

- a. **Visible Emission Line Coronagraph (VELC)**: It can peek as close as **1.05 solar radii**, a region never imaged by any solar telescope. It can thus give us more information about coronal mass ejection.
- b. **Solar Ultraviolet Imaging Telescope (SUIT)**: It will observe UV radiations from different zones of the solar atmosphere. It will help us to better understand the climate variation on earth.
- c. **Solar Low Energy X-Ray Spectrometer (SoLEXS)**
- d. **High Energy L1 Orbiting X-Ray Spectrometer (HEL1OS)**

» **In-Situ Payloads:**

- a. **Aditya Solar Particle Experiment (ASPEX)**: In-situ measurements of solar particles and ions.
 - b. **Plasma Analyzer Package for Aditya (PAPA)**
 - c. **Advanced Tri-axial High Resolution Digital Magnetometers**
- With the help of ASPEX, PAPA, and ATHRDM scientists can predict probable geomagnetic storms and better understand space weather dynamics.
 - **Conclusion:** Successfully placement of Aditya-L1 is a resounding vindication of India's investment in space science research, which can on the one hand spur fundamental enquiry of our cosmos and on the other generate knowledge of strong societal relevance.

2) XPoSat AND X-RAY POLARIMETRY

▫ **Why in news?**

- ISRO launched the XpoSat onboard a PSLV C58 flight on 1st Jan 2024.

▫ **Practice Question:**

- Discuss the scientific significance of XPoSat in advancing our understanding of high energy astrophysical phenomenon. How does it complement NASA's Imaging X-Ray Polarimetry Explorer (IXPE). [10 marks, 150 words]

▫ **More Details**

- XPoSat is a specialized science mission that studies the polarization of X-Rays in space. It was launched on board PSLV-C58 on 1st Jan 2024.
- **Need of studying polarization of X-Rays:** The mechanization of polarization of radiation gives away the nature of its source, including the strength and distribution of the magnetic field and the nature of radiation around it.
- XPoSAT has carried two scientific payloads in a low earth orbit:

- » The **Primary Payload (POLIX) (Polarimeter Instrument of X-Rays)** developed by RRI, measures the polarimetry parameters (degree and angle of polarization) in medium X-ray range (8-30 KeV) photons of astronomic origin.
 - » The **XSPECT (X-Ray Spectroscopy and Timing)**, developed by URSC, studies X-rays of energy 0.8-15 Kev and changes in the continuous emissions.
- **Need/Significance: Better understanding of the universe:** So far polarization studies were done on low energy radiations in optical and radio bands. XPOSAT by studying polarization in medium energy band (8-30 Ke) will be able to act as an excellent diagnostic tool to understand the emission process from astronomical sources. It will be:
- » **Unveiling the Mysteries of Black Holes:** It is one of the primary focus of XPoSat. By studying the polarization of X-rays emitted from the regions around the black holes, scientists can gain valuable information about geometry and strength of the magnetic field around black holes.
 - » **Better Understanding Neutron stars** which are remnants of massive stars.
 - » **More insights into high energy astrophysical processes** like particle acceleration, magnetic field amplification etc.
 - » **Verifying various models of Physics** with the help of Data from XpoSat.
 - » **Future Advancements** – the successful launch of XpoSat has paved the way for future more advanced launches leading to improved study of the universe.
- **International Trend in Space-Based X-Ray Polarimetry**
- » Internationally, space-based x-ray polarimetry is gaining importance.
 - The **Imaging X-Ray Polarimetry Explorer (IXPE)** mission, launched in 2021, represents NASA's inaugural space-based endeavor, focused on scrutinizing X-Ray Polarization across various celestial bodies.
 - **Note:** XPoSAT energy range of 8-30 keV for polarization measurement is complimentary to IXPE energy range of **2-8 KeV** (soft X-Ray band). Therefore, XPoSat and IXPE spacecrafts will collectively probe different emission mechanisms and physics for bright X-Ray sources. Their coordinated observation will provide a wide window in the energy range of 2-30 KeV for polarimetric observations for bright X-Ray sources.
- **In Conclusion**, XpoSAT is a landmark mission which is going to revolutionize our understanding of high energy astrophysics leading to new discoveries and advancement in the boundaries of human knowledge.

3) GAGANYAAN – THE HUMAN SPACE FLIGHT PROGRAM

- Example Question:

- Explain the significance of Gaganyaan Mission in India's future space program, S&T advancements, energy security, and strengthening of India's soft power. What are the key steps taken over the last few years to make the mission successful.
- Discuss the key significance of Gaganyaan Mission in India's Space Program [10 marks, 150 words]
- Evaluate the safety measures and risk mitigation strategy being implemented as part of Gaganyaan mission [10 marks, 150 words]

- Introduction

- **Gaganyaan project** is envisaged as a **technology demonstration project to showcase human spaceflight capability by India**. Under this, ISRO plans to launch a crew of 3 members in LEO for a 3-day mission and bring them back safely.
- **Test flights** (uncrewed) for the project are planned in 2024 and 2025. On success of these, the crewed flight is scheduled.

- Significance/ Need of HSP

- **First step towards future space programs** like having India's own space station and sending humans to moon and on other interplanetary mission. PM Modi has set the target of having a space station by 2035 and **landing an Indian on Moon by 2040**.
- **Resources outside earth:** Manned space programs are important to get control over resources. For e.g., the moon has enough helium-3 to power the entire globe.
 - » In fact, Dr K Sivan (former chair of ISRO) says that if colonies must be set up outside earth for the human civilization, how can India, one of the oldest civilizations, be left behind and waiting.
- **More R&D in space** – ISRO will be able to conduct micro-gravity experiment.
- **Advances in Science and Technology:** Mission will provide us opportunities to explore new technologies including push for indigenization, capacity building, and domestic manufacturing of new technologies.
- **Encouragement to our scientific community:** it will give self belief to our engineers, scientists etc.
- **Strengthen India's Soft power:** So far, only three countries USA, Russia and China have executed Human Spaceflight at their own. A successful mission would be ISRO's ultimate demonstration of capability.
- **Technological spin-offs will benefit other sectors.**

- » Space programs in the past have also resulted in hundreds of important spin-off technologies, including some breakthrough interventions in health and medicine sector like Magnetic Resonance Imaging (MRI), heart implants, and advanced water filtration.
- » Other examples include lithium-ion batteries developed by ISRO for rockets are now finding usage in commercial vehicles.
- » Fire Resistant Chemicals that ISRO will be developing for human flights may also be useful in petroleum industry, safeguarding workers from the fire.
- **Improvements in Higher Education** in the field of aeronautical engineering, aerospace engineering and physics.
- **Symbolism: Great power status** – Achievements in outer space are a marker of great power status.

- Key components of Human Space Program (HSP)

- **Human Space Flight Centre (HSFC)** – A body set up by ISRO as a coordinating body for Gaganyaan called the Human Space Flight Centre (HSFC).
- Development of Human rated launch vehicles for carrying crew safely to space.
 - » **All systems in LVM3 launch vehicles** are-reconfigured to meet human rating requirements and christened **Human Rated LVM3 (HLVM3)**. It will be capable of launching the orbital module to an intended LEO of 400 km.
 - » **HLVM3** also consist of Crew Escape System (CES) powered by a set of quick acting, high burn rate solid motors which ensure that Crew module along with crew is taken to safe distance in case of emergency either at launch pad or during ascent phase.
- **Orbital Module:** It is the object that LVM-3 rocket will launch and place in LEO. It will be orbiting earth and comprises of Crew Module (CM) and Service Module (SM). It is fit with adequate redundancy considering human safety.
 - » **CM** is the habitable space with Earth like environment in space for the crew. It can house upto three astronauts for a week. It includes:
 - Parachutes to slow its descent to the ground once it descends from orbit.
 - An **Environment Control and Life Support System (ECLSS)** to control the temperature, breathing environment, waste disposal, fire protection etc.
 - It also include a gynoid (feminine robot) named 'Vyomamitra' fit with sensors to track the effects of radiation and weightlessness, monitor capsule conditions, and sound alarms in the event of an impending emergency, aside from being able to perform some task.



- It has **double walled construction** consisting of pressurized metallic Inner Structure and unpressurised External Structure with Thermal Protection System (TPS).
 - » **SM** will be used for providing necessary support to CM while in orbit. It is an unpressurized structure containing thermal system, propulsion system, power systems, avionics systems and deployment mechanisms. The propulsion system here will raise the orbital module's altitude once it separates from the rocket and later to propel it back towards the earth.
 - **Precursor Missions** for demonstrating the technology preparedness levels before carrying out the actual mission. This includes Integrated Air Drop Test (IADT), Pad Abort Test (PAT), and Test Vehicle (TV) flights.
 - **Crew Training Facility established** in Bengaluru caters to Classroom training, Physical Fitness Training, Simulator Training and Flight suit training.
 - » **Training Modules** cover academic courses, Gaganyaan Flight Systems, Micro-Gravity familiarization through parabolic Flights, Aero-medical training, Recovery & Survival training etc. **Aero medical training**, Periodical Flying Practice and Yoga are also included as part of the training.
 - » **Note:** A shortlist of candidates was sent to Russia for advanced training.
 - **Capabilities for recovering astronauts safely.**
- **Other Steps taken so far:**
- **Space Borne Assistance and Knowledge Hub for Crew Interaction (SAKHI):** A multipurpose app developed by Vikram Sarabhai Space Centre that will help astronauts on Gaganyaan space flight mission carry out a range of tasks such as looking up vital technical information or communicating with one another. It will also monitor the health of astronauts, alert them about their dietary schedule etc. It will also help them stay connected with Earth. It will keep the crew connected with the onboard computer and ground-based stations, guaranteeing a seamless communication link.
 - **Pilots have been selected** and the identities of the four astronaut-designates, all IAF test pilots, were revealed in Feb 2024. The final crew for the mission will be chosen from among the four. **Prashanth Balakrishnan Nair, Ajit Krishnan, Angad Pratap and Shubhanshu Shukla** – are all Airforce pilot.
 - In 2018, the Union Cabinet had allocated Rs 10,000 crore for the program.
- **Conclusion – Class discussion**

4) BIRD FLU PANDEMIC

- **Why in news?**
 - » H5N1 (High Pathogenic Avian Influenza (HPAI)) bird flu has become widespread in wild birds worldwide and is causing outbreaks in poultry sector all across the world. It has also affected cattles across several states of USA, and for the first time, three cases of human infection in dairy farm workers were also reported (June 2024 - TH)
- **Practice Question:**
 - » Discuss the key challenges emerging globally due to the recent rise in cases of Avian Influenza. Suggest key measures to deal with these challenges [10 marks, 150 words]
- **Introduction:**
 - » Avian Influenza (Bird Flu) is caused by a type of influenza A viruses. Among various viruses, **only viruses of H5 and H7 subtype** are known to cause **Highly pathogenic Avian Influenza (HPAI)**. **H5N1** is the most common virus causing bird flu and its recent resurgence has presented significant challenges to humankind.
- **Key challenges:**
 - » **Public Health Risks:** While the current strain is not that easily transmitted to humans, but several recent cases in humans indicate the potential future risk.
 - As per WHO, between 2003 and 1st April 2024, 900 human infections have been reported across 23 countries and more than half of them were fatal.
 - Further, there is a grave risk of sustained transmission if the virus collects a few mutations while spreading among mammals.
 - » **Negative Economic Impact:** The loss of poultry sector globally has been in billions of dollars due culling of birds, trade restrictions and reduced demand from customers.
 - » **Biodiversity loss** - Several species, including in the isolated Antarctica continent, are facing challenges due to spread of bird flu. E.g. Emperor Penguin.
 - » **Potential to spread uncontrollably** - due to migratory pattern of wild birds across the world.
 - » **Increased public health expenditure** on the development of potential vaccine for cattles against bird flu.
 - For e.g. in May 2024, the US Department of Agriculture (USDA) issued a "request for information" from companies that have the capability to develop and produce cow vaccines against H5N1.
 - Scientists at University of Pennsylvania have developed an mRNA vaccine which have shown good results in pre-clinical trials.
- **Level of risk to humans:**
 - » The virus has spread from birds/animals to humans who were in close contact without right protective gear.
- **Steps already taken:**
 - » **Integrated disease surveillance Program (IDSP)** under the National Centre for Disease Control, Delhi, is supposed to keep tabs on the avian influenza in India.
- **Way Forward:**

- » **More investment in surveillance and early detection** including rapid diagnostic tools is crucial for swift response to bird flu outbreak.
 - » **Preventive Measures:**
 - **Wearing of protective gears** by people who work in close contact with livestock and birds.
 - Antiviral Tamiflu is being prescribed as a prophylaxis to people wherever H5N1 positive cases are found.
 - **Ensuring that people are only consuming pasteurized milk and well-cooked poultry meat and eggs.**
 - » **The concept of 'One Health'** needs to be prioritized and operationalized.
 - For e.g. Kerala is implementing a project as part of World Bank aided '**Rebuild Kerala**' project in four districts - Alappuzha, Pathanamthitta, Kottayam, and Idukki. Under this it has **established a community-based disease surveillance network**, with 2.5 lakh volunteers trained in the 'One Health' concept in these districts, who will report any unusual events or deaths of animals/birds in the locality so that early warning and early preventive or control measures can be taken.
 - » A system **to broadly survey/monitor wild birds arriving in India** needs to be developed. This has to be a long program for monitoring of wild bird movement on the lines of Project Tige.
 - » **International cooperation on vaccine development:** In the post COVID-19 world, a coordinated and focused effort towards vaccine development against bird flu for cattles and other animals can bear fruits much faster because of advancements in biotechnology.
- **Conclusion:**
- » The above measures can help the global community to effectively manage and mitigate challenges posed due to the recent resurgence of Avian influenza.

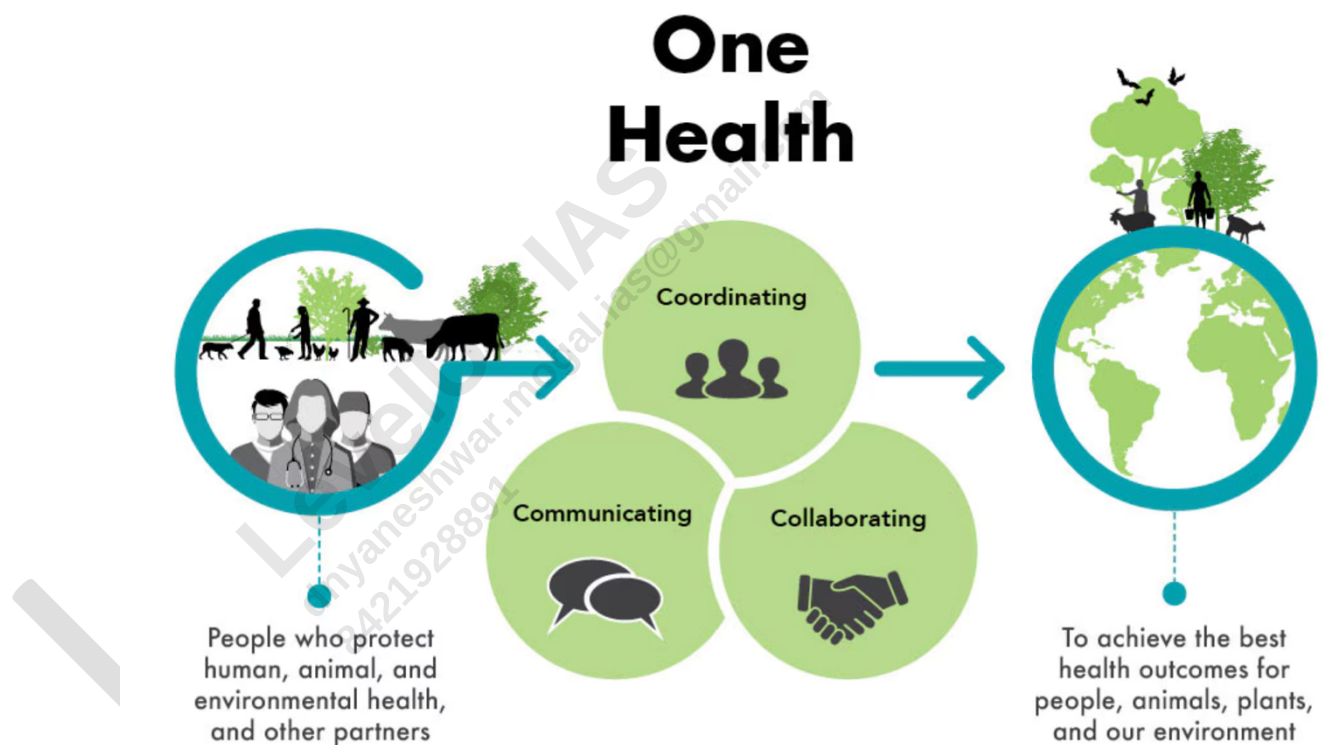
5) ONE HEALTH APPROACH

- **Practice Question:**
 - » India has of late been taking significant strides to deploy concepts and strategies rooted in the idea of 'One Health' to bolster the way it responds to health crisis. Evaluate the significance of One Health strategy in India's fight against zoonotic diseases [10 marks, 150 words]
- **Definition:**
 - » One Health is an approach that recognizes that health of people is closely connected to the health of animals and health of our shared environment. It is not a new concept and has been studied since ancient times. For instance, the writings of Hippocrates (460- 367 BCE) contemplated the relationships between public health and clean environment.
- **It is crucial for:**
 - » Controlling the outbreak of zoonotic diseases, improve food safety and security, reduce AMR resistance, protecting global health security, protecting biodiversity and environment, and ensuring better resource utilization by sharing expensive labs/resources.
- **Understanding the interconnection between human, animal, and environment health:**

- » **Human interaction with animals is increasing:** Human population is expanding, and more and more people are living in close contact with wild and domestic animals. This is leading to increased case of zoonotic diseases. E.g. Nipah, COVID-19 etc.
- » **Unsustainability in animal husbandry sector** harms human health. For e.g. overuse of antibiotics in animal husbandry is leading to development of anti-microbial resistance.
- » **Globalization** has led to increase in movement of people, animals and animal products. This leads to disease quickly spreading across borders and around the globe.
- » **Climate Change and Land use changes** is leading to disruption in environmental conditions which can provide new opportunities for diseases to pass to humans. For e.g. disease vectors like mosquitoes now cover more geographical area due to climate change.
- » **Environmental pollution** also exposes both humans and animals to several harmful substances leading to negative health impact.

- **How does a one Health Approach work?**

- » One Health approach is an effective way to fight health issues at the human-animal-environment interface. Successful public health intervention under this approach requires **coordination, communication, and collaboration** among professionals associated with human health (doctors, nurses etc.), animal health (veterinarians), environmental health (ecologists) and other relevant players like law enforcement agencies, policy makers, agriculture communities, pet owners etc.



- **Steps taken in India to promote One-Health Approach:**

- » Setting up of **National One Health Mission** was endorsed by PM's Science, Technology, and Innovation Advisory Council (PM-STIAC) in July 2022. Under this mission, various ministries (MoH&FW, Ministry of Animal Husbandry, Ministry of Science), departments (DBT, Department of Pharmaceuticals) and research institutions (CSIR, ICAR etc) have come together to shape the mission.

- **The goals** of the mission are to develop strategies for integrated disease surveillance, joint outbreak response; coordinated R&D; and ensuring seamless information sharing for better control of routine diseases as well as those of a pandemic nature.
 - The participating departments like DBT, CSIR are focused on **strong R&D** for the development of vaccines, therapeutics, and diagnostics.
 - Under the mission, a national network of **high-risk pathogen** (Biosafety level or BSL 3 and BSL 4) laboratories has been created.
 - These laboratories will serve to address the disease outbreak response better regardless of human, animal and environment sectors.
 - **Augmentation of epidemiology and data analytics capability**- Under this mission efforts are being made to apply AI and ML and disease modelling to address these issues and coordinate capacity building in epidemiology across sectors.
- » Foundation stone for **National Institute for One Health** has been laid in Nagpur in Dec 2022. it will anchor coordinating activities nationally and will also be the nodal agency to coordinate international activities cross the space of one health.
- **Some challenges of implementing One Health Approach in India:**
 - » **Institutional and policy fragmentation:**
 - Health is a state subject in India which limits the scope for coordination, collaboration and communication between states.
 - Similarly, Ministries like MoH&FW, MoA&FW, Ministry of Science etc. work in silos.
 - » **Limited infrastructure** - Lack of laboratory capacity, diagnostic tools hamper early detection and response to zoonotic diseases.
 - » **Limited Human resources** - Scarcity of health workers who understand the One Health approach also limits India's ability to achieve full potential of one Health approach.
 - » **Fragmented Data Collection system and lack of standardized reporting** also remain a challenge in the one health approach.
 - » **Issue with WHO's One Health Approach:**
 - WHO's One Health approach may undermine state sovereignty as it may impose legal obligation to share data on pathogens or genetic resources.
- **Conclusion1:** By promoting cooperation across all sectors, a One Health Approach can achieve the best health outcomes for people, animals, and plants in a shared environment.
- **Conclusion2:** One Health approach will further the goal of moving closer to the clarion call of 'One Earth, One Health' and 'Health for All'.

6) REFORMING POULTRY SECTOR IN INDIA

- **Current Context:** outbreak of H5N1 (already discussed in the above topics)
- **Key Problems caused by Poultry Sector:**
 - » **Unsafe conditions in Industrial livestock production:**
 - Crammed chicken in wire cages, or 'battery cages', in high densities create contaminated environment.

- These unsanitary conditions not only have detrimental effect on the welfare of animals, but also the health of those who consume the food derived, and on the people working at these facilities and residing in the vicinity. (e.g. increases chances of spread of Bird flu)
 - » **Air Quality and Waste Problem:**
 - **CPCB** has classified poultry units with more than 5,000 birds as a **polluting industry** that require compliance and regulatory consent to establish and operate.
 - Poultry sector creates odour, particulate matter, and other greenhouse gas emissions.
 - » **Antibiotics** are regularly given to birds as a **prophylactic** and as **growth promoters**. Several of these antibiotics are classified as critically important and highly important by the WHO and are sold to farmers for preventive use.
 - » **Heavily stocked animals in unsanitary conditions** -> detrimental effect on welfare of animals but also negative impact on those who consume the food derived from these animals.
 - **Violation of the Prevention of Cruelty to Animals (PCA) Act, 1960** – Keeping animals in intensive confinement constitutes a crime. Further, operational activities at these industrial facilities cause unnecessary pain and suffering to the animals because of mutilation, starvation, thirst, overcrowding, and other ill-treatment, which is also violation of PCA.
 - » **Other challenges:** The sector also suffers from lack of access to finance, lack of skilled manpower, market volatility, fluctuating feed prices etc.
- **Way Forward:**
- » **Recommendations of Law Commission of India** in its 269th report:
 - Non-therapeutic antibiotic given to poultry causes antibiotic resistance.
 - More open, cleaner, and ventilated living spaces are likely to cause less need of antibiotics in animals.
 - This will also make their eggs and meat safer to eat.
 - They laid guidelines as per existing laws and international best practices for animal care, waste management etc.
 - But, when government (MoA&FW) came with Draft Rules for the egg industry in 2019, they were weak and tokenization. The rules need to follow law commission recommendations.
 - » **Strict regulation** which adheres to Prevention of Cruelty to Animals (PCA) Act, 1960 of the poultry sectors.
 - » **Developing biofertilizer facilities** around poultry clusters.
 - » **Infrastructure improvement** in poultry sector:
 - Investments in cold storage and supply chain will be crucial for ensuring quality of food products. Animal Husbandry Infrastructure Development Fund which has been extended upto 2025-26 should also prioritize poultry sector.
 - » **Awareness among consumers** about health impact of food from properly managed poultry farms
 - » Other challenges of poultry sector can be solved by increasing access to finance and insurance, developing training programs to enhance to skills of poultry workers, and developing alternative feed source.
- **Conclusion:**
- The above reforms are urgently needed for poultry sector not only for the welfare of animals but also because animal welfare is closely interlinked with public health, ecosystem health and biodiversity conservation.

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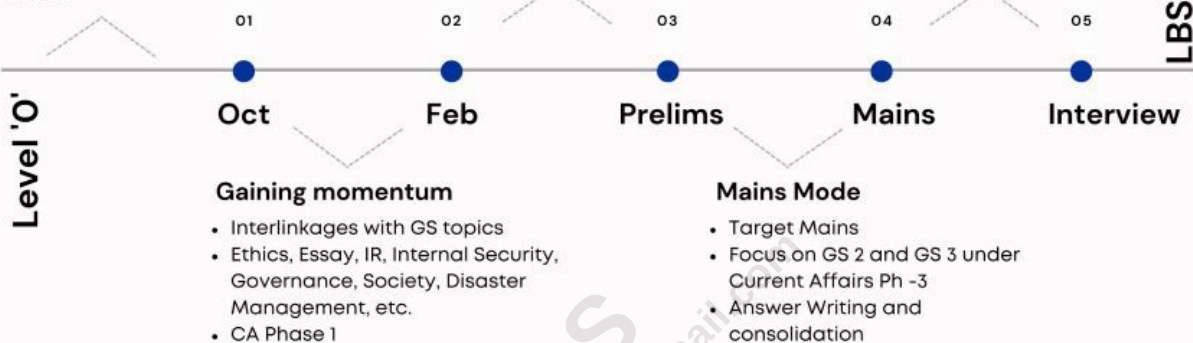
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